

This Patient Group Direction (PGD) is intended only for registered healthcare professionals who have been named and authorised by their organisation to practice under it.

Patient Group Direction
for the administration of Hydroxocobalamin for the treatment of Vitamin B12 Deficiency

Summary of NICE CKS Guidelines for Low Vitamin B12

Overview

- Vitamin B12 deficiency can cause **megaloblastic anaemia**, **neurological symptoms**, and **non-specific symptoms** (e.g. fatigue).
- Causes include **pernicious anaemia**, **malabsorption**, **dietary deficiency**, and **medications** (e.g. metformin, PPIs).

Symptoms

General:

- Fatigue
- Weakness
- Dyspnoea
- Pale or jaundiced skin

Neurological:

- Paraesthesia (numbness/tingling)
- Peripheral neuropathy
- Cognitive changes
- Ataxia

Haematological:

- Macrocytic anaemia
- Glossitis
- Low white blood cells/platelets (in severe cases)

Diagnosis

- **Check serum B12 levels** if:
 - Symptoms suggestive of deficiency.
 - Macrocytic anaemia on FBC.
 - Risk factors present (e.g. vegan diet, elderly, GI surgery).
- Also assess:
 - **Folate and ferritin** (other deficiencies may coexist).
 - **Intrinsic factor antibodies** (for pernicious anaemia).
 - **MMA and homocysteine** (specialist settings) if B12 borderline.

Note: Normal B12 levels can still occur in symptomatic patients — clinical judgement is key.

Causes

- **Pernicious anaemia** (autoimmune destruction of intrinsic factor).
 - **Dietary deficiency** (especially in vegans).
 - **Malabsorption** (e.g. post-gastrectomy, Crohn's disease).
 - **Medications** (e.g. metformin, PPIs, H2 blockers).
 - **Alcohol misuse, elderly, GI disorders.**
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Management

If symptomatic or confirmed B12 deficiency:

1. Hydroxocobalamin injections (first-line in the UK):

- **1 mg IM every other day for 2 weeks** (initial correction phase).
- Then:
 - **Every 3 months for life** if due to **pernicious anaemia or irreversible cause**.
 - **Stop after correction** if reversible (e.g. dietary), and re-evaluate oral intake.

2. Oral B12 (cyanocobalamin):

- For **dietary deficiency only** where absorption is intact.
 - Dose: 50–150 mcg daily (higher doses may be used off-label).
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Monitoring and Follow-Up

- Check symptom improvement after 4–6 weeks.
- Repeat FBC after initial treatment if anaemia was present.
- Monitor neurological symptoms closely — may take **months to resolve**.
- Lifelong monitoring if cause is permanent.

When to Refer

- Diagnostic uncertainty.
- Neurological symptoms not improving.
- Borderline B12 with normal haematology but persisting symptoms.
- Suspected malabsorption or coexisting conditions (e.g. coeliac disease).

Patient Group Direction

for the supply/administration of

Hydroxocobalamin 1mg

for the treatment of

Vitamin B12 deficiency

Healthcare professionals covered and training

	Healthcare professionals allowed to work under this PGD and requirements
Qualifications and Registration	● Pharmacist registered and practicing with the GPhC ● Pharmacy technician registered and practicing with the GPhC
Training and assessment	● Pharmacists and Pharmacy Technicians must have completed training relevant to this condition to use this PGD. This training will be documented and overseen by the Get Real Health team. ● All users must be familiar with the SPC for the products as well as BNF advice and any applicable national guidelines ● All users must previously have used PGD's to supply medication ● Must work in compliance with the SOPs of their own employer ● All users must have appropriate indemnity in place to cover this service
Further training and responsibilities	● All users must be up to date with CPD requirements and appraisal ● All users must be up to date with MHRA and safety alerts with regards to medical products ● All users must understand the concepts of capacity and consent and be aware of the Mental Capacity Act 2005.

PGD Indication	Hydroxocobalamin is indicated for the treatment of vitamin B12 deficiency, including pernicious anaemia, malabsorption, or dietary deficiency. It supports neurological function, red blood cell formation, and DNA synthesis.
Inclusion criteria	- Confirmed diagnosis of vitamin B12 deficiency (e.g., blood tests, clinical symptoms). - Adults aged 18 years or older. - No contraindications to hydroxocobalamin. - Informed consent obtained. - Suitable for intramuscular injections in a non-acute setting.
Exclusion criteria	- Known hypersensitivity to hydroxocobalamin or cobalamin derivatives. - Active infection at proposed injection site. - Known hereditary problems of cobalamin metabolism. - Severe thrombocytopenia or bleeding disorders unless managed appropriately. - Individuals with Leber's disease (risk of optic atrophy).
Cautions	- Monitor potassium levels during initiation in severe deficiency (risk of hypokalaemia). - Inform patients that urine may turn red post-injection. - Rare anaphylactic reactions may occur – observe after administration. - Use with caution in individuals with history of allergies or asthma.
Actions if patient is excluded or declines treatment	Advise on alternative treatment options and how these can be accessed. Document any advice given and the decision reached. Inform or refer to the GP as appropriate

Details of the medicine

Name, form and strength of medicine	Hydroxocobalamin 1mg
Legal category	POM
Storage	Store below 25°C. Protect from light.
Route/method of administration	Intramuscular injection.
Dose and frequency	Initial correction: 1 mg intramuscularly three times a week for 2 weeks. Maintenance: 1 mg every 2 to 3 months depending on underlying cause and response.
Quantity to be administered and/or supplied	1 vial of 1mg
Maximum or minimum treatment period	Initial correction followed by long-term maintenance as clinically indicated.
Adverse effects	Injection site reactions, dizziness, nausea, headache, rash, and rarely

	anaphylaxis. Urine may appear reddish post-injection. Refer to SmPC.
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Yellow card reporting	Report suspected adverse effects via https://yellowcard.mhra.gov.uk and inform the GP if appropriate.
Records to be kept	Record the following information: ● that valid informed consent was given ● name of individual, address, date of birth and GP ● name of healthcare practitioner ● name and brand of medication ● date of supply dose, form and route quantity supplied ● advice given, including advice given if excluded or declines treatment ● details of any adverse drug reactions and actions taken supplied via PGD Records should be signed and dated. All records should be clear, legible and contemporaneous. Electronic records can be made. A record of all individuals receiving treatment under this PGD should also be kept for audit purposes as follows: ● Adults aged 18 years and over - keep records for 8 years ● Children aged under 18 years -keep records until the 25th birthday (if the patient was 17 years when treatment was finished, keep records until the 26th birthday)

Patient information

Written information to be given to patient or carer	Supply the patient information leaflet (PIL) provided with the medication.
Follow-up advice to be given to patient or carer	To seek medical advice if symptoms worsen rapidly or significantly, do not improve in 3–4 weeks, or they become systemically very unwell.

Key references

- NICE Medicines Practice Guideline 2 (MPG2): Patient Group Directions. Published March 2017 <https://www.nice.org.uk/guidance/mpg2>
- NICE MPG2 Patient group directions: competency framework for health professionals using patient group directions. Updated March 2017 <https://www.nice.org.uk/guidance/mpg2/resources>
- NICE CKS Guidance
- British National Formulary (BNF) Key information on the selection, prescribing, dispensing and administration of medicines [BNF \(British National Formulary\) | NICE](#)
- Summary of Product Characteristics

Agreement to practise by the Registered Healthcare Professional

By signing this PGD you are confirming that you agree to its contents and that you will work within it. PGDs do not remove inherent professional obligations or accountability. It is the

responsibility of each healthcare professional to practise only within the bounds of their own competence and professional code of conduct.

Name and address of pharmacy premises / clinic premises to which this PGD relates			
Name of healthcare professional	Job title	Registration number	Signature
Valid from date		Expiry date	

PGD adoption and authorisation by the employer/clinical lead

The employer has ensured that the person using this PGD, has the knowledge and skills to safely provide the service which will encompass the supply or administration of a medication or medicine.

This section must be signed by the superintendent or clinical lead responsible for this service.

Name of superintendent / clinical lead	Job title & name of organisation	Signature	Date

Name of professional group signatory	Job title & name of organisation	Signature	Date
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Signed on behalf of Get Real Health

Name	Qualification	Signature	Date
Nitin Shori	Medical Doctor GMC: 6047293		25/8/25
Chris Pilkington	Pharmacist GPhC: 2046322		25/8/25

Change history

Version number	Change details	Date
001	Development & issue of new PGD	25 th June 2025